PIXANT Manual

Rapid and accurate multi-phenotype imputation for millions of individuals

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Table of Contents

PIXANT Manual ..... - 1 -
1 Brief introduction ..... - 4 -
2 PIXANT function ..... - 4 -
2.1 Description ..... - 4 -
2.2 Usage ..... - 4 -
2.3 Arguments ..... - 5 -
2.4 Value ..... - 5 -
3 PIXANT.eval function ..... - 6 -
3.1 Description ..... - 6 -
3.2 Usage ..... - 6 -
3.3 Arguments ..... - 6 -
3.4 Value ..... - 6 -
4 prodNA function ..... - 6 -
4.1 Description ..... - 6 -
4.2 Usage ..... - 7 -
4.3 Arguments ..... - 7 -
5 simG function ..... - 7 -
5.1 Description ..... - 7 -
5.2 Usage ..... - 7 -
5.3 Arguments ..... - 7 -
6 simPhen function ..... - 7 -
6.1 Description ..... - 7 -
6.2 Usage ..... - 8 -
6.3 Arguments ..... - 8 -
7 imputeUnivariate function ..... - 8 -
7.1 Description ..... - 8 -
7.2 Usage ..... - 8 -
7.3 Arguments ..... - 8 -
8 PhenAdj function ..... - 8 -
7.1 Description ..... - 9 -
7.2 Usage ..... - 9 -
7.3 Arguments ..... - 9 -

7.4 Value ..... - 9 -
8 sampleScore function ..... - 9 -
8.1 Description ..... - 9 -
8.2 Usage ..... - 9 -
8.3 Arguments ..... - 9 -
8.4 Value ..... - 10 -
9 Build in data ..... - 10 -
9.1 Running build-in data ..... - 10 -
10 Code availability ..... - 11 -
11 How to access help ..... - 11 -

1 Brief introduction

Deep phenotype datasets can enhance the power of genetic analysis such as genome-wide association study (GWAS), but recurrence of missing phenotypes compromises the potentials of such resources. Here we address the central issue of missing phenotypes in studies with any level of relatedness between phenotypes. we propose a multi-phenotype imputation method that is scalable to large data over a million individuals. We call our method PIXANT (multi-**p**henotype **i**mputation method based on mi**x**ed f**a**st ra**n**dom fores**t**). PIXANT models nonlinear and linear effects across multi-phenotype correlation and higher-order interactions between predictive factors, and brings out unbiased imputation of much higher accuracy than state-of-the-art methods. Tested in a dataset of n=20,000 individuals and p=30 phenotypes, PIXANT is ~24.45 times faster and uses only about one ten thousandth memory compared with PHENIX(***Nature Genetics*** 2016 (48):466–472). Moreover, real data set analysis and biologically plausible results suggest that our method imputation can uncover new true positive results.

2 PIXANT function

PIXANT	Imputing missing values by using the mixed fast random forest.

2.1 Description

Imputing missing values by using the mixed fast random forest. The PIXANT method comprises two parts: 1) the estimation part: estimating parameters under the null model and applying the likelihood to correct for linear and nonlinear effects; 2) the imputation part: performing imputation for the missing values.

2.2 Usage

```
PIXANT(xmis, MaxIterations = 20, MaxIterations0 = 20, num.trees = 100, mtry =
        floor(sqrt(ncol(xmis))), initialLinearEffects = 0, ErrorTolerance = 0.001,
        indPhen = NULL, missingSize = 500, seed = 123, replace = TRUE,
        decreasing = TRUE, verbose = TRUE, sampsize = NULL, max.depth =
        NULL, xtrue = NA)
```

2.3 Arguments

xmis	A vector, matrix or data frame with missing values.
MaxIterations	Stop after how many iterations (default = 20).
MaxIterations0	The maximum iteration times of mixed fast random forest (default = 20).
num.trees	How many trees are grown in the mixed fast random forest (default = 100).
mtry	How many variables should be tried randomly at each node.
initialLinearEffects	The initial values for linear effects (default = 0).
ErrorTolerance	The tolerance for log-likelihood (default = 0.001).
indPhen	The columns of the target phenotype, and used only for cross-validation operation (default = NULL).
missingSize	The missing values size of settings in the target phenotype, and must be less than number of observed values of the target phenotype (default = 500).
seed	Random seed. Default is 123, which generates the seed from R. Set to 0 to ignore the R seed.
replace	(boolean) If TRUE bootstrap sampling (with replacements) is performed, else subsampling (without replacements).
decreasing	(boolean) If TRUE the columns are sorted with decreasing amount of missing values.
verbose	(boolean) If TRUE then PIXANT returns error estimates, runtime and if available true error during iterations.
sampsize	List of size(s) of sample to draw.
max.depth	Maximal tree depth. A value of NULL or 0 (the default) corresponds to unlimited depth, 1 to tree stumps (1 split per tree).
xtrue	The complete true data (a vector, matrix or data frame).

2.4 Value

Return a list, the list contains:

ximp	The imputed data (a vector, matrix or data frame).
accuracy	Imputation accuracy of target phenotype.
value	The missing data of settings in the target phenotype, including sample index, observed values and imputed values (a data frame).

r2	The r square of fitting between observed values and imputed values in the target phenotype.
pValue	The p value of fitting between observed values and imputed values in the target phenotype.

3 PIXANT.eval function

PIXANT.eval	PIXANT evaluation indicator.

3.1 Description

The function calculates correlation coefficient between imputed values and their true hidden values.

3.2 Usage

PIXANT.eval(ximp, xmis, xtrue)

3.3 Arguments

ximp	The imputed data (a vector, matrix or data frame).
xmis	The data with missing values.
xtrue	The complete true data.

3.4 Value

Return the correlation coefficient between the real values and the imputed values.

4 prodNA function

prodNA	Produce missing values.

4.1 Description

This R script contains the function to produce missing values in a given and data set completely at random.

4.2 Usage

prodNA(x, noNA, seed)

4.3 Arguments

x	A vector, matrix or data frame.
noNA	Proportion of missing values to add to x. In case x is a data frame, noNA can also be a vector of probabilities per column or a named vector (see examples).
seed	An integer seed.

5 simG function

simG	Simulated Genome Relationship Matrix.

5.1 Description

The function simulates the construction of a genome relational matrix.

5.2 Usage

simG(N, k, fam_size)

5.3 Arguments

N	The number of individuals and must be a positive integer.
k	Coefficient of kinship and the value ranges from 0 to 1.
fam_size	The size of the family, fam_size must be a positive integer and must divide N.

6 simPhen function

simPhen	Simulated phenotype.

6.1 Description

This function simulates the phenotypes for individuals.

6.2 Usage

simPhen(N = N, P = P, K = G, h2 = rep(0.6, P), B, E)

6.3 Arguments

N	The number of individuals.
P	The number of phenotypes.
K	A genome relational matrix.
h2	The heritability of each phenotype in individuals.
B	Genetic covariance. (allow the missing)
E	Environmental or residual covariance. (allow the missing)

7 imputeUnivariate function

imputeUnivariate	Univariate Imputation.

7.1 Description

Fills missing values of a vector, matrix or data frame by sampling with replacement from the non-missing values. For data frames, this sampling is done within column.

7.2 Usage

imputeUnivariate(xmis, v = NULL, seed = NULL)

7.3 Arguments

xmis	A vector, matrix or data frame with missing values.
v	A character vector of column names to impute (only relevant if x is a data frame). The default NULL imputes all columns.
seed	An integer seed.

8 PhenAdj function

PhenAdj	Adjusted phenotypic values.

7.1 Description

Adjusted phenotypic values base on covariates.

7.2 Usage

PhenAdj(Phen, Cov)

7.3 Arguments

Phen	Phenotype file. The missing values should be denoted by NA.
Cov	A matrix of covariates. Each row is a sample and each column corresponds to one covariate. For example, age, gender.

7.4 Value

Return adjusted phenotype file.

8 sampleScore function

sampleScore	QC.

8.1 Description

Estimating the SC (Sample Score) for each phenotype of each individual.

8.2 Usage

sampleScore(Phen, use="pairwise",method="spearman",adjust="fdr",alpha=.05)

8.3 Arguments

Phen	Phenotype file. The missing values should be denoted by NA.
use	use="pairwise" is the default value and will do pairwise deletion of cases. use="complete" will select just complete cases.
method	method="pearson" is the default value. The alternatives to

	be passed to cor are "spearman" and "kendall". These last two are much slower, particularly for big data sets.
adjust	What adjustment for multiple tests should be used? ("holm", "hochberg", "hommel", "bonferroni", "BH", "BY", "fdr", "none"). See p.adjust for details about why to use "holm" rather than "bonferroni").
alpha	alpha level of confidence intervals.

8.4 Value

Return the SC (Sample Score) for each phenotype of each individual.

9 Build in data

An example dataset 'ukb' is the UK Biobank data set. The data including a data frame (10000 * 8), each row represents 8 phenotypes information for an individual. The 'ukb' can be loaded with data(ukb).

```
ukb                          10000 obs. of 8 variables
  ID_23111: num -13.07 -2.48 3.32 8.66 13.6 ...
  ID_23115: num -12.47 -1.39 3.76 8.39 13.1 ...
  ID_30020: num 0.589 -1.494 -0.533 0.747 -1.558 ...
  ID_30840: num -2.57 -1.65 3.47 -3.34 -3.55 ...
  ID_30850: num 6.53 -6.1 -4.59 -4.49 -5.72 ...
  ID_50 : num 9.7983 0.0104 -0.5577 -4.4106 -2.5829 ..
  ID_30660: num -0.766 -0.534 0.994 -0.641 NA ...
  ID_30690: num 0.0599 0.2644 -0.6888 2.9311 -1.5519 .
```

9.1 Running build-in data

```
library("PIXANT")
data(ukb)
ukb.mis <- ukb
PIXANT.imp <- PIXANT(ukb.mis, MaxIterations = 20, MaxIterations0 = 20,
                     num.trees = 100, mtry = floor(sqrt(ncol(ukb.mis))),
                     initialLinearEffects = 0, ErrorTolerance = 0.001,
                     indPhen = 7, missingSize = 500, seed = 123, replace =
                     TRUE, decreasing = TRUE, verbose = TRUE, sampsize
                     = NULL, max.depth = NULL, xtrue = NA)
```

10 Code availability

The source code of PIXANT is freely available.

11 How to access help

If users have any bugs or issues or any suggestions are available, feel free to contact:

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